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BIO302 Current Midterm spring 2023 solve BY MAS All Rounder

1. Chargaff's rule?

Answer:

Chargaff's rule states that in DNA, the amounts of adenine (A) are same to the amounts of thymine (T), and the quantities of guanine (G) are same to the amounts of cytosine (C). This rule became important because it furnished critical evidence for the bottom pairing in DNA and helped in elucidating the structure of DNA.

2. Characteristics of genetic material?

Answer:

The characteristics of genetic material include:
- It consists of and stores genetic records.
- It undergoes replication to skip at the genetic statistics to the subsequent generation.
- It is capable of undergoing mutations, which make contributions to genetic version.

• It may be expressed and transcribed into purposeful molecules which include RNA and proteins.
• It is subject to regulation and interacts with other cellular components to carry out various features.

3. Telomerase and its function?

Answer?

Telomerase is an enzyme that is involved in retaining the length and integrity of telomeres, which can be protective structures discovered on the ends of chromosomes. Telomerase adds repetitive nucleotide sequences to the telomeres, stopping their shortening at some point of DNA replication. Its predominant characteristic is to counteract the slow shortening of telomeres that takes place with every spherical of DNA replication, thereby preventing genomic instability and cellular senescence.

4. Chemical composition of protein?

Answer:

Proteins are composed of amino acids, which might be related together through peptide bonds. The chemical composition of a protein includes carbon (C), hydrogen (H), oxygen (O), nitrogen (N), and occasionally sulfur (S). Amino acids comprise an amino organization (-NH2), a carboxyl group (-COOH), and a facet chain (R-institution) that varies among different amino acids. The sequence and arrangement of amino acids decide the shape and function of a protein.

5. Why Cisplatin is cytotoxic and act as chemotherapeutic agent?

Answer:

Cisplatin is cytotoxic and acts as a chemotherapeutic agent because it forms covalent bonds with DNA, leading to the formation of DNA adducts. These adducts because DNA crosslinking, DNA strand breaks, and interfere with DNA replication and transcription. By inducing DNA harm, cisplatin disrupts the cellular cycle and triggers apoptosis, ultimately leading to the loss of life of cancer cells.

6. Role of 8 Oxo guanine?

Answer:

8-Oxo guanine is a changed form of guanine which could end result from oxidative harm to DNA. It is a mutagenic lesion and can cause the mispairing of nucleotides in the course of DNA replication, inflicting G:C to T:A transversions. Eight-Oxo guanine is diagnosed and repaired through unique enzymes concerned in base excision repair (BER) to maintain genomic integrity.

7. End of replication error?

Answer

End of replication error refers to the unfinished replication of the very give up of a linear DNA molecule at some point of DNA replication. Due to the mechanism of DNA replication, the final segment of the lagging strand, known as the telomere, cannot be fully replicated. As a result, it leads to the sluggish shortening of telomeres with every spherical of replication. Telomerase enables to counteract this shortening by way of including repetitive sequences to the telomeres.

8. Name three proteins that catalyzes the direct alkylation?

Answer:

Three proteins that catalyze direct alkylation are:

• DNA alkyltransferase (additionally referred to as O6-methylguanine-DNA methyltransferase or MGMT): It upkeep O6-alkylguanine adducts.

• AlkB homologs (ALKBH): They repair alkylated bases, together with 1-methyladenine and three-methylcytosine.

• Ada protein: It maintenance O6-methylguanine and O4-methylthymine adducts.

9. 2. How can one differentiate between pentose of ribose and deoxyribose?

Answer:

Ribose and deoxyribose are exceptional sugars located in nucleotides. Ribose is the sugar factor of ribonucleotides, which can be the constructing blocks of RNA. Deoxyribose is the sugar aspect of deoxyribose nucleotides, which might be the constructing blocks of DNA. The principal distinction between them is that ribose has a hydroxyl organization (-OH) attached to its 2' carbon, at the same time as deoxyribose has a hydrogen atom (-H) connected to its 2' carbon.

10. Name the origin of DNA, where alkylation occurs?

Answer:

Alkylation can arise at numerous sites in DNA, however one distinguished web page is the O6 function of guanine. Alkylation at this function can result in DNA damage. One beginning of alkylation in DNA is publicity to alkylating marketers present within the surroundings or positive chemotherapeutic capsules, along with cisplatin.

11. 4. Watson and Crick work?

Answer:

Watson and Crick's paintings refers to the groundbreaking discovery of the shape of DNA. In 1953, James Watson and Francis Crick proposed the double helix model of DNA, which defined the molecule as a twisted ladder with complementary base pairing and antiparallel strands.

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Their work became based on the existing statistics and insights from other scientists, which includes Rosalind Franklin's X-ray diffraction pix.

12. Discriminative amino acids?

Answer:

Discriminative amino acids, additionally known as variant or non-conserved amino acids, are amino acid residues that fluctuate among extraordinary proteins or protein sequences. These residues contribute to the precise houses, structure, and function of every protein.

13. Explain nucleotide excision repair (NER) in five points?

Answer:

Nucleotide excision repair (NER) is a DNA repair mechanism that operates to put off a extensive variety of DNA lesions, including cumbersome adducts, pyrimidine dimers, and different styles of DNA damage. It involves several steps:

14. Scientist's k experiments?

Answer:

Recognition and preliminary harm verification.

15. NER or BER?

Answer:

NER (Nucleotide Excision Repair) and BER (Base Excision Repair) are each DNA repair mechanisms however with one-of-a-kind specificities:

- NER is involved in putting off bulky DNA lesions, inclusive of UV-brought on pyrimidine dimers and chemical adducts.
- BER broadly speaking offers with the repair of damaged or changed bases, consisting of oxidative DNA harm and small chemical adjustments.

16. Characteristic of hereditary material?

Answer:

The traits of hereditary material encompass:

- It consists of and transmits genetic facts from one technology to the next.
- It can undergo replication to make certain correct inheritance.
- It can undergo mutation, main to genetic variant.
- It determines the developments and traits of an organism.

17. Component of DNA?

Answer:

The components of DNA are nucleotides. A nucleotide is composed of a sugar molecule (deoxyribose in DNA), a phosphate group, and one of the four nitrogenous bases: adenine (A), thymine (T), cytosine (C), or guanine (G).

18. How chromatin exist in histone protein?

Answer:

Chromatin exists as a complex of DNA and histone proteins. Histone proteins are chargeable for packaging DNA right into a relatively organized and compact shape. DNA wraps round histone octamers to shape nucleosomes, which can be the basic devices of chromatin. This prepared packing of DNA with histones allows adjust gene expression and protects DNA from damage.

19. Semi conservative model of DNA?

Answer:

The semi-conservative version of DNA replication states that during DNA replication, every new DNA molecule contains one strand from the authentic DNA molecule (the template strand) and one newly synthesized complementary strand. This model changed into proposed via Watson and Crick and has been experimentally verified.

20. DNA repair mechanism?

Answer:

DNA restore mechanisms are critical for preserving the integrity of the genome. Several DNA restore mechanisms exist, together with base excision repair (BER), nucleotide excision repair (NER), mismatch restore (MMR), homologous recombination (HR), and non-homologous quit joining (NHEJ). These mechanisms correct diverse sorts of DNA damage, making sure correct DNA replication and cellular viability.

21. Base Excision Repair?

Answer:

Base Excision Repair (BER) is a DNA repair mechanism that particularly objectives broken or changed bases in DNA. It entails several steps:
Recognition and elimination of the damaged base by means of a selected DNA glycosylase enzyme.
Incision of the DNA spine by using an endonuclease enzyme.
Removal of the damaged DNA segment.
DNA synthesis to fill the gap.
Ligation to seal the repaired DNA strand.

22. Diff between nucleotide and nucleoside?

Answer:

Nucleotide and nucleoside are related terms in biochemistry:
• A nucleotide includes a nitrogenous base (adenine, thymine, cytosine, or guanine), a sugar molecule (ribose in RNA or deoxyribose in DNA), and a phosphate group.
• A nucleoside consists of a nitrogenous base and a sugar molecule (ribose or deoxyribose) but lacks the phosphate group.

23. Coenzyme with example?

Answer:

Coenzyme is a non-protein molecule that assists enzymes in wearing out unique reactions. It regularly acts as a provider of chemical companies or electrons. An example of a coenzyme is nicotinamide adenine dinucleotide (NAD+), which performs a essential role in redox reactions, wearing electrons from one response to another.

24. Mitochondria k functions?

Answer:

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Mitochondria have numerous essential capabilities in cells:
• They generate strength in the form of ATP via oxidative phosphorylation.
• They play a position in mobile metabolism and the manufacturing of metabolites.
• They are worried in calcium homeostasis and apoptosis.
• They have their own DNA and are chargeable for the synthesis of some of their proteins.

25. Difference between RNA and DNA?

Answer:

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The most important variations between RNA and DNA are as follows:
• RNA has the sugar ribose, while DNA has the sugar deoxyribose.
• RNA is generally single-stranded, while DNA is double-stranded.
• RNA uses the nitrogenous base uracil (U), even as DNA uses thymine (T).
• RNA is concerned in numerous mobile methods, which includes protein synthesis, at the same time as DNA consists of genetic records and serves because the blueprint for protein synthesis

26. PCR applications?

Answer:

PCR (Polymerase Chain Reaction) is a method used in molecular biology for amplifying particular DNA sequences. It has diverse packages, including:
• DNA cloning and genetic engineering.
• DNA sequencing.
• Diagnosing genetic diseases.

• Forensic evaluation.
• Environmental and evolutionary research.

27. Role of cellular membrane?

Answer:

The cellular membrane, also known as the plasma membrane, serves several roles:
• It acts as a selective barrier, regulating the shipping of molecules into and out of the mobile.
• It maintains the mobile's inner surroundings by using controlling the access and go out of ions and nutrients.
• It enables cell-cellular communique and sign transduction.
• It gives structural guide and shape to the cellular

28. What is telomeres and its functions??

Answer:

Telomeres are regions of repetitive DNA sequences on the ends of chromosomes. Their capabilities include:
• Protection of chromosomes from degradation and fusion.
• Maintenance of chromosome stability for the duration of replication and cell division.
• Prevention of DNA damage reaction and mobile senescence.
• Regulation of mobile lifespan and getting older.

29. What is nucleotide excision repair and give five steps?

Answer:

Nucleotide excision restore is a DNA restore mechanism that is accountable for getting rid of cumbersome DNA lesions, together with those precipitated with the aid of ultraviolet (UV) light. Here are the 5 steps concerned in nucleotide excision restore:

30. Why C-N bond of peptide cannot rotate freely?

Answer:

Recognition of DNA damage: The first step is the popularity of DNA damage by a complex of proteins referred to as the harm popularity complex. This complicated scans the DNA for abnormalities or distortions within the DNA structure.

31. How phosphodiester bond is formed??

Answer:

31. Phosphodiester bonds are fashioned all through the system of DNA and RNA synthesis, in which they be part of nucleotides together to form a polymer chain. The formation of a phosphodiester bond involves a condensation response among the 3'-hydroxyl organization of 1 nucleotide and the 5'-phosphate institution of some other nucleotide. The reaction outcomes within the formation of a phosphodiester linkage and the release of a water molecule. This

system is catalyzed by using enzymes known as DNA polymerases or RNA polymerases, depending on whether DNA or RNA synthesis is taking place. The successive formation of phosphodiester bonds between nucleotides creates a backbone of alternating sugar and phosphate companies, with the nucleotide bases extending from the sugar moieties. This backbone forms the structural framework of DNA and RNA molecules and is critical for their balance and feature.

32. How would harsher and chase learn weather genes were made of protein or DNA?

Answer:

In their test to determine whether genes had been manufactured from protein or DNA, Alfred Hershey and Martha Chase carried out a examine called the Hershey-Chase test in 1952. They used bacteriophages, which might be viruses that infect bacteria, as their version machine.
In the experiment, Hershey and Chase classified the DNA and proteins of bacteriophages with one of a kind radioactive isotopes. They used radioactive sulfur-35 (35S) to label the proteins and radioactive phosphorus-32 (32P) to label the DNA. The radioactive isotopes served as markers that might be detected and tracked.
The researchers allowed the categorized bacteriophages to infect bacteria. After contamination, they used a blender and a centrifuge to split the viral coats (containing proteins) from the inflamed microorganism. They observed that maximum of the radioactive phosphorus-32 (DNA marker) was located inside the bacterial cells, at the same time as the radioactive sulfur-35 (protein marker) remained inside the viral coats.
These findings led Hershey and Chase to conclude that during the infection system, the genetic fabric of the bacteriophage, which includes the information needed for viral replication, is DNA, no longer protein. The test furnished robust proof supporting DNA as the hereditary fabric.
The Hershey-Chase experiment played a vast function in confirming that DNA, rather than proteins, is the molecule responsible for sporting genetic data. This discovery laid the foundation for the information of DNA's valuable role in genetics and paved the manner for in addition studies in molecular biology.

33. Silent features of DNA polymerase?

Answer:

DNA polymerase is a critical enzyme concerned in DNA replication and repair approaches. It has several wonderful functions that make a contribution to its function:
a. DNA synthesis: DNA polymerase catalyzes the synthesis of a new DNA strand by adding nucleotides in a complementary manner to the existing DNA template strand.
B. Proofreading interest: Many DNA polymerases own a proofreading or exonuclease hobby. This lets in them to discover and accurate mistakes that occur for the duration of DNA replication by means of eliminating incorrectly incorporated nucleotides.
C. High processivity: DNA polymerase well-knownshows excessive processivity, which means it can synthesize long stretches of DNA without dissociating from the template strand. This is done via the association of DNA polymerase with other proteins, which includes sliding clamps

that hold the enzyme connected to the DNA.
D. Three'-5' exonuclease activity: Some DNA polymerases possess a 3'-five' exonuclease pastime that permits them to get rid of nucleotides from the three' give up of a DNA strand. This interest is worried in proofreading and DNA repair procedures.
E. Template-directed synthesis: DNA polymerase synthesizes the brand new DNA strand in a template-directed way that means it makes use of the present DNA strand as a template to decide the sequence of the newly synthesized strand.
These features of DNA polymerase make certain the correct replication and restore of DNA, playing a essential position in retaining the integrity of the genome. Different types of DNA polymerases exist with versions of their specific features and properties.

34. Contribution of Molecular biology in the field of agriculture?

Answer:

Molecular biology has made significant contributions to the field of agriculture. Some of the key contributions encompass:
a. Genetic engineering of crops: Molecular biology techniques, together with recombinant DNA era, have enabled the genetic amendment of vegetation to introduce desirable tendencies. This includes developments like multiplied yield, pest resistance, herbicide tolerance, and progressed dietary content material.
B. Development of genetically modified organisms (GMOs): Molecular biology has facilitated the creation of genetically changed organisms, such as genetically changed plants. GMOs were evolved to improve crop productiveness, reduce crop losses because of pests and sicknesses, and decorate nutritional value.
C. Crop development and breeding: Molecular biology equipment, together with marker-assisted choice (MAS) and genomic choice, have revolutionized crop breeding programs. These strategies allow breeders to discover and pick out flowers with favored tendencies greater effectively and as it should be, main to the improvement of improved crop types.
D. Disease diagnostics and control: Molecular biology strategies, along with polymerase chain reaction (PCR) and DNA sequencing, have substantially progressed the diagnosis and control of plant sicknesses. These techniques allow speedy and correct detection of pathogens, identification of resistance genes, and monitoring of disorder outbreaks.
E. Understanding plant genetics and physiology: Molecular biology studies has deepened our know-how of plant genetics and physiological approaches. This knowledge allows in elucidating the molecular mechanisms underlying plant boom, improvement, and responses to environmental stresses. It offers insights for developing strategies to optimize crop production and enhance strain tolerance.
Overall, molecular biology has played a pivotal role in advancing agricultural practices, main to elevated crop yields, superior meals fine, and more sustainable farming methods.

35. PCD structure and its function?

Answer:

PCD stands for Programmed Cell Death, additionally known as apoptosis. PCD is a quite regulated system by way of which cells undergo managed self-destruction. It is a essential

mechanism in diverse organic techniques, which includes development, tissue homeostasis, and immune reaction.
The shape and characteristic of PCD contain several key additives:
a. Morphological adjustments: During PCD, cells undergo feature morphological modifications, inclusive of cellular shrinkage, chromatin condensation, nuclear fragmentation, and the formation of apoptotic bodies. These adjustments assist within the green elimination of loss of life cells without inflicting irritation.
B. Activation of caspases: Caspases are proteases that play a crucial role in executing PCD. Activation of caspases initiates a cascade of events main to cellular demise. Caspases cleave specific cellular proteins, resulting in dismantling of the mobile and activation of apoptotic pathways.
C. Apoptotic pathways: PCD can be initiated thru two foremost pathways: the extrinsic pathway and the intrinsic pathway. The extrinsic pathway is activated by external indicators, consisting of binding of loss of life ligands to demise receptors at the cellular surface. The intrinsic pathway is prompted by using internal alerts, consisting of DNA damage or cell pressure, main to the release of seasoned-apoptotic elements from mitochondria.
D. Regulation of PCD: PCD is tightly regulated by a balance between seasoned-apoptotic and anti-apoptotic alerts. Various molecules, along with Bcl-2 circle of relatives proteins, play crucial roles in regulating PCD by means of both promoting or inhibiting the apoptotic technique.
E. Signaling and clearance: PCD includes signaling mechanisms to apprehend and take away death cells. Phagocytic cells understand uncovered "eat me" signals on apoptotic cells and engulf them, stopping the discharge of cell contents and retaining tissue homeostasis.
PCD is essential for regular development, keeping tissue integrity, and removing broken or doubtlessly dangerous cells. Dysregulation of PCD is associated with diverse illnesses, which includes most cancers and neurodegenerative disorders. Understanding the shape and function of PCD has big implications for biomedical studies and therapeutic interventions.

36. Name Radiations that damage DNA?

Answer:

Several styles of radiations are regarded to harm DNA. These encompass:
a. Ionizing radiation: Ionizing radiation, which include X-rays, gamma rays, and excessive-strength debris (e.G., alpha and beta particles), can without delay harm DNA by means of causing breaks in the DNA strands. These breaks can result in chromosomal aberrations and genetic mutations.
B. Ultraviolet (UV) radiation: UV radiation, especially UV-B and UV-C, can induce DNA damage by forming covalent bonds between adjacent pyrimidine bases (thymine dimers). This sort of DNA harm interferes with ordinary DNA replication and transcription strategies.
C. Radioactive decay merchandise: Certain radioactive materials, which include radon gasoline and its decay merchandise, emit alpha particles that can harm DNA by means of without delay ionizing atoms within the DNA molecule.
D. Environmental toxins: Some environmental toxins, like certain chemicals and pollution, can generate reactive oxygen species (ROS) that result in oxidative harm to DNA. ROS can motive DNA base modifications, strand breaks, and different types of DNA lesions.

These radiation-precipitated DNA damages will have numerous outcomes, along with genetic mutations, chromosomal rearrangements, and accelerated danger of cancer development. Cells have DNA repair mechanisms to counteract the damaging effects of radiation, but immoderate or extended publicity to radiation can weigh down those repair processes. It is critical to minimize exposure to dangerous radiation and take suitable precautions to defend DNA integrity.

37. Which is the simplest amino acid residue?

Answer:

The best amino acid residue is glycine. Glycine is the smallest and most effective amino acid, which includes only a hydrogen atom as its side chain (R-organization). It is considered the most effective as it lacks a facet chain that consists of extra atoms or practical organizations determined in different amino acids. Glycine performs critical roles in protein shape and feature and is worried in various biochemical procedures inside the frame. It is classified as a nonpolar, aliphatic amino acid.

38. How many types of ribonucleic acids are present in body?

Answer:

Several forms of ribonucleic acids (RNAs) are gift within the frame. The essential types of RNAs include:

a. Messenger RNA (mRNA): mRNA includes the genetic information from DNA to the ribosomes, in which it serves as a template for protein synthesis. It incorporates the instructions for the amino acid collection of a protein.

B. Transfer RNA (tRNA): tRNA molecules are accountable for transferring specific amino acids to the ribosomes during protein synthesis. Each tRNA molecule acknowledges a selected codon at the mRNA and contains the corresponding amino acid.

C. Ribosomal RNA (rRNA): rRNA is a structural thing of ribosomes, which are the mobile machinery answerable for protein synthesis. It facilitates within the binding of mRNA and tRNA and catalyzes the formation of peptide bonds among amino acids.

D. Small nuclear RNA (snRNA): snRNAs are worried inside the splicing of pre-mRNA molecules, which removes introns and joins exons to supply the mature mRNA molecules which might be translated into proteins.

E. MicroRNA (miRNA): miRNAs are small RNA molecules that alter gene expression by way of binding to mRNA molecules and inhibiting their translation or selling their degradation. They play critical roles in publish-transcriptional gene law.

F. Small interfering RNA (siRNA): siRNAs are double-stranded RNA molecules which can silence particular genes through targeting and degrading complementary mRNA molecules.

These are a number of the principal sorts of RNAs determined in the body, each with its specific function in gene expression, protein synthesis, and regulatory techniques.

39. Which RNA contains anticodon?

Answer:

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Transfer RNA (tRNA) carries the anticodon. The anticodon is a series of 3 nucleotides at the tRNA molecule this is complementary to a specific codon at the mRNA during protein synthesis. The anticodon guarantees the correct pairing among the mRNA codon and the corresponding amino acid carried via the tRNA.
The complementary binding between the anticodon and the codon lets in the tRNA to supply the ideal amino acid to the growing polypeptide chain throughout translation. For instance, if the mRNA codon is "AUG" (coding for the amino acid methionine), the tRNA anticodon could be "UAC" (complementary to "AUG").
The presence of the anticodon on tRNA molecules ensures the fidelity and specificity of the genetic code translation, allowing the appropriate amino acids to be included into the growing polypeptide chain in line with the mRNA collection.

40. Which strand replicates in forward direction?

Answer;

The strand that replicates inside the ahead path is the main strand. During DNA replication, the leading strand is synthesized continuously within the 5' to three' route, which is the equal route as the replication fork movement. The leading strand is capable of be synthesized continuously due to the fact its orientation allows for the addition of nucleotides in a smooth and uninterrupted way. In contrast, the lagging strand is synthesized discontinuously in short fragments known as Okazaki fragments, which might be later joined collectively.

41. Which enzyme breaks the dsDNA and passes same amount of DNA from the break?

Answer:

The enzyme that breaks the double-stranded DNA and passes the same quantity of DNA from the ruin is DNA topoisomerase. DNA topoisomerase is answerable for relieving the anxiety and supercoiling that takes place beforehand of the replication fork for the duration of DNA replication. It achieves this by using developing temporary breaks inside the DNA strands, permitting the passage of the replication machinery. Once the DNA has been surpassed via, the breaks are resealed, and the DNA strands are intact. DNA topoisomerase is crucial for the clean progression of the replication method. [1]

42. What is the structure of CPD and also write its function?

Answer:

Clouting pyrimidine dimers (CPDs) are DNA lesions that form while two adjoining pyrimidine bases, usually thymine or cytosine, come to be covalently connected. CPDs are most generally fashioned via exposure to ultraviolet (UV) radiation. Their shape involves the formation of a covalent bond between two neighboring pyrimidine bases, resulting inside the bending or kinking of the DNA helix. CPDs can disrupt the regular shape and function of DNA and intervene with procedures along with DNA replication and transcription.
The characteristic of CPDs is to function a marker for DNA harm and trigger DNA repair mechanisms. Cells have specialized repair pathways, consisting of nucleotide excision restore

(NER), which understand and dispose of CPDs from the DNA molecule. NER includes excising the damaged DNA section and replacing it with a newly synthesized section the usage of the complementary strand as a template. The restore of CPDs is critical for preserving the integrity of the genome and stopping mutations that may cause sicknesses, consisting of pores and skin cancer.

43. What are the psi and phi values in alpha helix? Also write their values?

Answer:

The phi (ϕ) and psi (ψ) values constitute the torsional angles of the peptide backbone in an alpha helix secondary shape. These values describe the rotations around the C-N and C-C bonds, respectively, between the amino acid residues in the helix.
In an alpha helix, the phi attitude is approximately -60 ranges, even as the psi attitude is about -45 tiers. These values make contributions to the characteristic coiling of the alpha helix shape and allow for the hydrogen bonding between the amino acid residues alongside the helix.
It's vital to word that those values can vary barely depending at the unique amino acid residues present within the alpha helix. However, the overall phi and psi values cited above constitute the standard tiers located in alpha helices.

44. What is the reaction of DNA helicase in replication?

Answer:

DNA helicase plays a essential function in DNA replication. Its foremost function is to unwind and separate the double-stranded DNA molecule on the replication fork. It does this with the aid of breaking the hydrogen bonds between the complementary nucleotide base pairs, correctly "unzipping" the DNA helix.
Once the DNA helix is unwound and separated, DNA helicase also helps to stabilize the unwound DNA strands, preventing them from reannealing. This lets in the unmarried-stranded DNA template to be accessible for replication with the aid of DNA polymerase.
The hobby of DNA helicase is ATP-based, which means it requires strength from ATP hydrolysis to carry out its unwinding function. ATP binding and hydrolysis provide the strength essential for DNA helicase to transport alongside the DNA strand and separate the 2 strands.

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